

Quiz (Habanero)

Open Ended Questions (FRQ)

Q9. The temperature T in a region is given by $T(x) = x^2 - 4$, where x is the distance from a heat source. Suppose $x = 2s + 1$, where s is the time in hours. Evaluate $(T \circ \text{circ} x)(s)$ at $s = 2$. Show all steps and explain your reasoning.

Q10. In an ecosystem, the population P of a species is affected by the rainfall R . Suppose $P(R) = 2R + 1$ and $R(x) = x^2$, where x is the temperature. Determine $(P \circ \text{circ} R)(x)$ and explain its significance in the context of the ecosystem. Provide examples.

Chef's Tips

- Read each question carefully
- Check for domain restrictions
- Label all diagrams and graphs